

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: T.Y. BTECH.

Branch: Mechanical Engineering

Semester: V

Subject Code & Name: BTMEC504_19 Theory of Machines- II

Max Marks: 60

Date: 24-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The effect of increasing the diameter of the driving pulley in a belt drive system is: a. Increase the speed of the driven pulley b. Decrease the speed of the driven pulley c. Increase the power transmission d. No effect on the driven pulley speed		1
2	In a V-belt drive, the angle of the V-shaped groove helps to: a. Increase the tension in the belt b. Reduce slipping c. Increase the diameter of the pulleys d. Decrease the efficiency		1
3	The module is the reciprocal of..... a. Diametral pitch b. circular pitch c. pitch diameter d. none of the mentioned		1
4	The size of a gear is usually specified by a. pressure angle b. circular pitch c. Diametral pitch d. pitch circle diameter		1
5	When the axes of first and last gear are co-axial, then gear train is known as a. simple gear train b. compound gear train c. reverted gear train d. Epicyclic gear train		1
6	The type of gears used to connect two non-parallel and non-intersecting shafts is a. Spur gear b. Helical gear c. Bevel gear d. Spiral gear		1
7	A governor is said to be hunting, if the speed of the engine a. remains constant at the mean speed b. is above the mean speed c. is below the mean speed d. fluctuates continuously above and below the mean speed		1
8	The axis of rotation of the gyroscope's spinning wheel is called the _____		1

	a. Precession axis	b. Input axis	c. Spin axis	d. Output axis		
9	In aeroplanes, gyroscopic couple is most significant during:					1
	a. Straight flight	b. Banking	c. Rapid pitch changes (take-off/landing)	d. Engine shutdown		
10	For occurrence of free torsional vibration which of the condition is necessary?					1
	a. Rotors moving in same direction	b. Rotors having same frequency	c. Rotors having different frequency	d. Rotors rotate in the same sense		
11	The factor which affects the critical speed of a shaft is					1
	a. Diameter of the disc	b. Span of the shaft	c. Eccentricity	d. All of the mentioned		
12	The speed at which the shaft runs so that the additional deflection from the axis of rotation of the shaft becomes infinite, is known as					1
	a. Whirling speed	b. Rotational speed	c. Stabilizing speed	d. Reciprocating speed		
Q. 2	Solve the following.					12
A)	A flat belt drive transmits 35 kW from a pulley of 1.5 m effective diameter running at 300 rpm. The angle of contact is spread over 11/24 of the circumference, and the coefficient of friction between the belt and pulley surface is 0.3. Determine the width of the belt required, considering centrifugal tension.					6
B)	Why are rope drives suitable for long-distance power transmission? Explain with reference to their efficiency and design.					6
Q.3	Solve the following.					12
A)	Explain the concept of involute gear tooth profile. Why is it preferred over cycloidal profiles?					6
B)	A pinion with 25 teeth drives a gear with 75 teeth. The pinion rotates at 1200 rpm and transmits 200 Nm torque. Find: a) Speed of the gear b) Torque on the gear c) Power transmitted by the gear pair					6
Q. 4	Solve Any Two of the following.					12
A)	Describe a compound gear train. How does it improve velocity ratio in limited space?					6
B)	A flywheel of moment of inertia 50 kg-m ² runs at 200 rpm. If the speed fluctuates between 180 rpm and 220 rpm during the cycle, find the energy stored in the flywheel.					6
C)	Explain the working of a differential gear using a simple epicyclic gear train. Why is it used in vehicles?					6

Q.5	Solve Any Two of the following.		12
A)	Define and discuss the terms: sensitivity, stability, hunting, and isochronism in governors.		6
B)	Discuss the function of a flywheel and how it differs from that of a governor.		6
C)	A gyroscope has a rotor of mass 5 kg and radius of gyration 0.2 m, rotating at 3000 rpm. The ship in which it is mounted turns with an angular velocity of 0.1 rad/s. Calculate the gyroscopic couple acting on the ship.		6
Q. 6	Solve Any Two of the following.		12
A)	Discuss the effects of gyroscopic couple on ships, aeroplanes, and vehicles, with examples.		6
B)	A mass of 10 kg is attached to a spring with stiffness $k=2000\text{N/m}$. Determine: a) Natural frequency in Hz b) Natural angular frequency in rad/s.		6
C)	Explain the concept of torsional vibration in rotating shafts. Discuss its causes and effects on mechanical systems.		6
	*** End ***		